Quantitative text analysis – Essex Summer School

Comparing representations: sparse vectors, dense vectors, embeddings, and context

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Today's class

- What does security mean in parliamentary speech?
- Use that question to compare sparse counts, dense document vectors, static word embeddings, and contextual embeddings
- When are embeddings useful for a substantive social science question. Validation.
- Lab session 3

“Security” in parliamentary discourse

> "... the strength of our deterrent ...
> proud of our armed forces ..."

Keir Starmer, Labour, Defence and Security, 25 February 2025.

> "... good for the country's energy security,
> good for our economy and good for jobs."

Rishi Sunak, Conservative, Economic Update, 3 February 2022.

Research question: when MPs talk about security what do they talk about exactly?

From question to evidence

Document-level evidence

- How often do MPs use security-related words?
- For which agendas, parties, or decades do they appear the most?

Contextual evidence

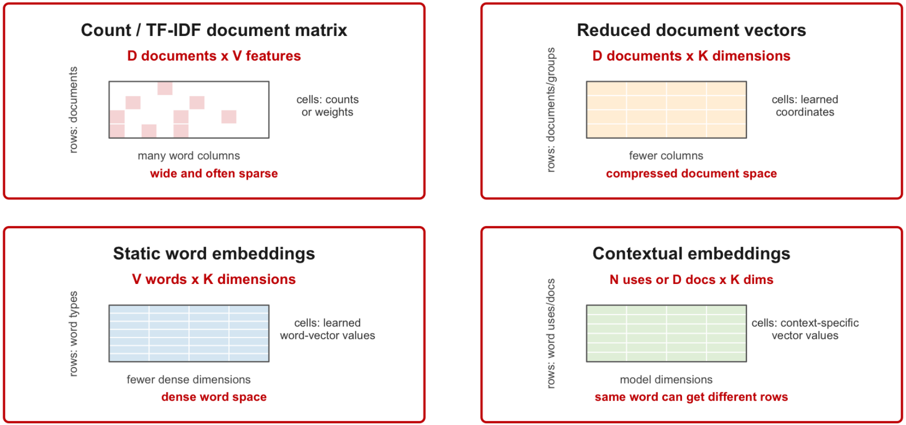
- Which words appear near `security`?
- Is `security` used in similar ways across speeches?

To address such questions, we need to think about representation, an object we create from text so that we can compare texts or words systematically.

Four representations

Representation	Object We Compare	Question
count vectors	speeches or grouped speeches	which words appear, and how often?
reduced document vectors	speeches or grouped speeches	can broad word-use patterns be summarized?
static embeddings	words	which words are used in similar surroundings?
contextual embeddings	word uses or whole speeches	does meaning change with the sentence?

What representations look like



D = documents; V = vocabulary; K = vector dimensions; N = word uses. Width and height are schematic, not exact counts.

The BoW representation

A document-feature matrix is often the first object we build.

- rows are speeches
- columns are features
- cells count word occurrences

excerpt	security	deterrent	forces	energy	economy	jobs
Starmer	0	1	1	0	0	0
Sunak	1	0	0	1	1	1

Counts answer prevalence questions: which terms are common, and where do they appear?

From sparse counts to dense document vectors

After bag-of-words counting, each document is represented by a long vector the size of the vocabulary.

- This is transparent: every column is a word or feature
- But it is also sparse: most documents have zeroes for most terms

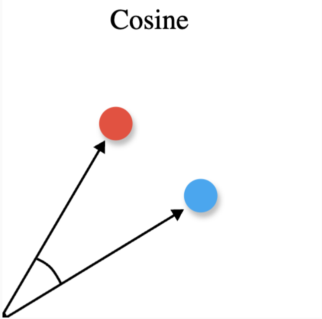
SVD / LSA compresses this matrix into fewer dimensions

$$\text{documents} \times \text{features} \rightarrow \text{documents} \times \text{dimensions}$$

- Each new dimension is a weighted combination of many terms
- Documents with similar word-count profiles receive similar coordinates
- The dimensions are learned summaries; you need to interpret what they mean.

Comparing vectors

Once text becomes vectors, we can compare rows or columns.



Cosine similarity

- Cosine similarity asks whether two vectors point in a similar direction
- This is useful when the relative pattern matters more than the absolute size of the vector
- In embedding models, cosine similarity is usually the default starting point
- Many other similarity and distance metrics exist, but we focus on cosine today

Figure: Maarten Grootendorst, `https://www.maartengrootendorst.com`.

Cosine similarity: a numeric example

Suppose two short speeches are represented by three binary features.

speech vector	`security`	`forces`	`energy`
defence speech	1	1	0
energy speech	1	0	1

Let $a = [1, 1, 0]$ and $b = [1, 0, 1]$.

Step	Calculation	Value
dot product	$(1)(1) + (1)(0) + (0)(1)$	1
length of a	$\sqrt{1^2 + 1^2 + 0^2}$	$\sqrt{2}$
length of b	$\sqrt{1^2 + 0^2 + 1^2}$	$\sqrt{2}$
cosine similarity	$1/(\sqrt{2} \times \sqrt{2})$	0.50

Intuition: the speeches share `security`, but each also contains one different word. Their cosine similarity is therefore moderate rather than high.

Why cosine similarity?

Cosine similarity is useful when we care about the direction of two vectors rather than their raw size.

- Two speeches can have different lengths but similar proportions of terms
- Two words can be represented by dense embedding vectors with similar patterns across dimensions
- The result is easy to interpret: higher values indicate more similar directions

Caution: cosine similarity is not a substitute for interpretation. It tells us which vectors are close; we still need to return to the text and ask whether the pattern makes substantive sense.

From bag of words to embeddings

Bag-of-words counts give us a transparent baseline.

- We can see which words appear
- We can explain exactly how the representation was built

But what if we a more interested in the meanings that `security` gets attached to in parliamentary language?

Rodman (2020) treats this as a problem of meaning in context and argues that it requires us to move is from counting a concept to studying the words and issues around it.

Embeddings as a next step

Word embeddings represent words as real-valued vectors, learned from the words that appear around them.

Rodriguez and Spirling (2022) highlight two common uses:

Use	What We Do With the Vectors	Example in This Lecture
features	feed embeddings into another model	classify speeches using dense vectors
measures	study the vectors themselves	ask what words are closest to `security`

Today we mostly use embeddings as measures.

What BoW sees, what embeddings see

Bag of words

```
security  -> [1, 0, 0, 0, ...]
deterrent -> [0, 1, 0, 0, ...]
energy    -> [0, 0, 1, 0, ...]
jobs      -> [0, 0, 0, 1, ...]
```

- each feature is separate
- similarity must be learned later

Illustrative numbers only.

Word embeddings

```
security  -> [ 0.12, -0.31, 0.44, ...]
deterrent -> [ 0.16, -0.28, 0.39, ...]
energy    -> [-0.05,  0.22, 0.18, ...]
jobs      -> [-0.09,  0.19, 0.21, ...]
```

- vectors are dense
- similarity is partly encoded

Intuition behind word embeddings

"You shall know a word by the company it keeps"

Firth (1957)

- Scan many speeches and record the words that appear around each term
- Words used in similar surroundings receive similar vectors
- The window determines what counts as around: a five-word window asks which words appear close to each other

tokenized speeches $\rightarrow$ word windows $\rightarrow$ dense vectors

Rodman (2020): from frequency to meaning

Rodman's starting point is that political concepts do not have fixed meanings.

- A word can become attached to different issues, actors, and conflicts
- So a frequency trend is only the start of the analysis
- We then ask what appears around the concept, and whether that context changes

Step	Applied to `security`
choose a concept	`security`
inspect its context	is it closer to `forces`, `energy`, or `jobs`?
compare settings	parties, agendas, decades
validate	return to speeches, KWIC examples, and substantive expectations

For us: word distances can become evidence about how actors connect concepts in political language, but only after returning to the texts.

Canonical static embedding approaches

All three approaches start from the same intuition: words are informative because of the company they keep.

Approach	Training Intuition	What To Remember
GloVe	count how often words appear near one another, then compress those patterns into vectors	starts from a global co-occurrence table
word2vec	learn vectors that help predict nearby words from a target word, or the target word from nearby words	treats meaning as a prediction problem
FastText	learn vectors from words and their character pieces	helps with rare words, morphology, and related word forms

Rodriguez & Spirling (2022): applied choices

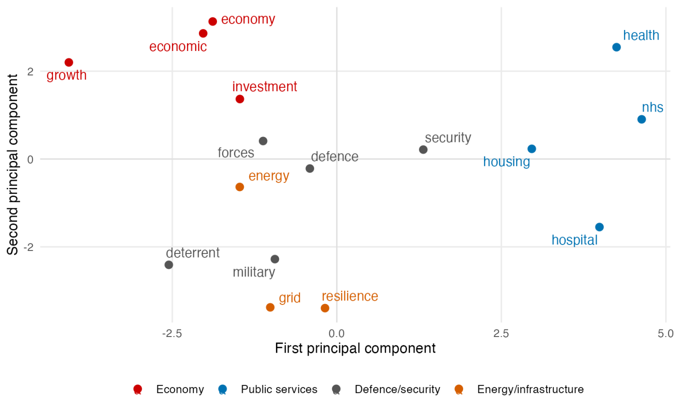
Rodriguez and Spirling's practical warning is that embeddings are useful, but not automatic. Each modelling choice changes the evidence we create.

Choice	Why It Matters for `security`
window size	narrow windows capture immediate phrasing; wider windows capture broader topic
vector dimensions	too few dimensions may flatten differences; too many may capture noise
local vs pretrained	HoC vectors capture parliamentary usage; pretrained vectors bring outside language patterns
model stability	nearest neighbours should not depend entirely on one random run
validation	close words should make sense when we return to the speeches

Visualizing word vectors

Think of words as points in a high-dimensional space.

- Words close together are used in similar contexts
- Distance is usually measured with cosine similarity
- A 2D plot is useful for intuition, but it is only a projection



PCA projection of selected GloVe word vectors from the HoC sample.

Contextual vector representations

Static embeddings give one vector per word type.

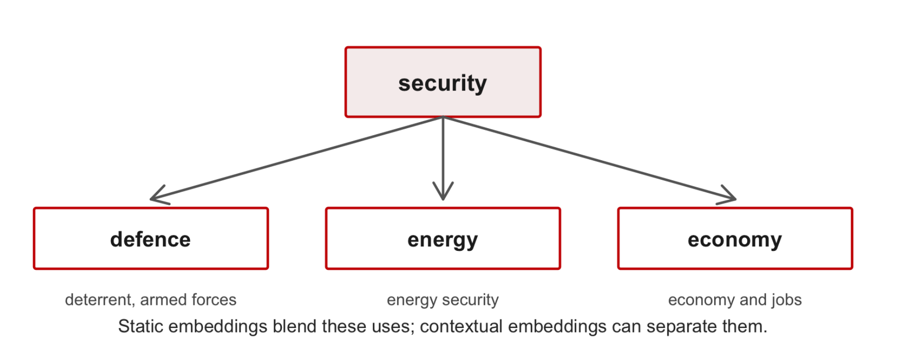
Transformer-based models create contextual vectors.

Representation	Example	What Changes?
static word vector	GloVe vector for `security`	one vector for the word type
contextual token vector	BERT vector for `security`	vector changes with sentence context
document embedding	embedding for a whole speech	one vector summarising a text

This is a bridge to Days 9 and 10: transfer learning, zero-shot classification, and LLM-assisted coding.

Same word, different contexts

The word `security` can be used in different ways:



What do embeddings encode?

Nearest neighbours can reflect more than semantic meaning.

Source of Similarity	Example Risk
parliamentary procedure	`hon`, `member`, `friend` dominate contexts
time period	older and newer vocabularies mix unevenly
rhetorical style	terms reflect style, not concept meaning
preprocessing	trimming or stopwords change the vocabulary
corpus imbalance	frequent topics shape the embedding space

Validation

Embeddings are powerful exploratory tools, but they are still measurements.

- Do the nearest neighbours recover a relationship we can defend?
- Are results similar if we change reasonable modelling choices?
- Do KWIC examples and original speeches support the interpretation?
- Are we measuring concept meaning, or something else such as procedure, period, or rhetorical style?